VIETNAM NATIONAL UNIVERSITY - HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Programming for Artificial Intelligence and Data Science (CO3135)

Assignment Submission

House Price Prediction

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Contents

1 Group Information & Member Roles **2**

2 Chosen Data & Justification **3**
2.1 Dataset Description 3
2.2 Feature Meanings 3
2.3 Justification 3

3 Work Progress Overview **5**

4 Exploratory Data Analysis (EDA) **6**
4.1 Data Quality Assessment 6
4.2 Analytical Dataset 9
4.3 Distributions and Relationships in the Analytical Dataset 10
4.4 Summary 13

1 Group Information & Member Roles

The registered group will work together on all assignments in this course. The responsibilities for this phase are divided as follows:

- **Member 1: Võ Nhật Tân**
  - Role: xxx.
- **Member 2: Ngân Phạm**
  - Role: xxx.
- **Member 3: Minh Thiện**
  - Role: xxx.
- **Member 4: Chi Phan**
  - Role: xxx.

2 Chosen Data & Justification

2.1 Dataset Description

This project addresses house price prediction for the Vietnamese market. The selected data source is the public dataset *Tinix Vietnam Real Estate Listings 2025–2026* [Tin26]. According to the published dataset card, the collection contains exactly 3,500,744 sale and rental listings published between June 2025 and March 2026, after a strict property-type filter, and covers all 63 provinces and cities of Vietnam. The listings are distributed as ten Parquet shards, are written primarily in Vietnamese, and are released under the CC BY-NC 4.0 licence for non-commercial academic use.

2.2 Feature Meanings

Table 1: Feature meanings from the published dataset card [Tin26].

Column	Type	Description (published)
`name`	string	Listing title (HTML stripped, phone numbers redacted)
`description`	string	Full listing description (HTML stripped, phone numbers redacted)
`property_type_name`	string	Property category (e.g., Căn hộ chung cư, Nhà, Đất, ...)
`province_name`	string	Province / City name (63 provinces)
`district_name`	string	District (Quận / Huyện)
`ward_name`	string	Ward (Phường / Xã)
`street_name`	string	Street name
`project_name`	string	Real estate development project name (if applicable)
`price`	float64	Listed price in VND (Vietnamese Dong). `null` if not disclosed.
`area`	float64	Total floor/land area in m^2
`floor_count`	float64	Number of floors in the building
`frontage_width`	float64	Frontage width in meters
`house_depth`	float64	Depth of the property in meters
`road_width`	float64	Width of the road in front of the property (meters)
`bedroom_count`	float64	Number of bedrooms
`bathroom_count`	float64	Number of bathrooms
`house_direction`	string	Cardinal facing direction (Đông, Tây, Nam, Bắc, Đông-Bắc, ...)
`balcony_direction`	string	Cardinal facing direction of the balcony
`published_at`	string	ISO 8601 datetime the listing was published

2.3 Justification

The dataset is suitable for the prediction task for three reasons.

Scale. The dataset contains over 3.5 million records, which is large enough to support stable exploratory analysis and supervised learning across regions and property types.

Schema. The schema separates a clear numeric target (`price`) from predictors describing location (`province_name`, `district_name`, `ward_name`, `street_name`, `project_name`), property category (`property_type_name`), size and layout (`area`, `floor_count`, `frontage_width`, `house_depth`, `road_width`, `bedroom_count`, `bathroom_count`), orientation (`house_direction`, `balcony_direction`), textual content (`name`, `description`), and listing time (`published_at`).

National coverage. The dataset covers all 63 provinces and cities of Vietnam, which makes it feasible to study regional price patterns and to train models that generalise beyond a single city.

The published definitions also indicate limitations that call for an EDA stage before modelling: `price` may be undisclosed (`null`), some structural fields are often incomplete for particular property types, and the collection includes both sale and rental listings, with no variable that distinguishes them. These issues are examined in Section 4, where an analytical subset of priced listings is constructed for subsequent analysis and model training.

3 Work Progress Overview

4 Exploratory Data Analysis (EDA)

This section assesses the quality of the raw listings, constructs the analytical dataset of priced listings, and describes the distributions of price and area. The analysis was carried out in two notebooks, one on data quality and cleaning and one on price analysis. Each listing is treated as a property offered for sale, and `price` is interpreted as the asking price in Vietnamese dong (VND). The results therefore cover only listings that disclose a price, and some rental listings may remain (see Section 2).

4.1 Data Quality Assessment

Schema and data types. The raw data contain 3,500,744 listings and the 19 variables described in Table 1. All expected columns were present, with no additional columns. Eight variables are numeric (`price`, `area`, `floor_count`, `frontage_width`, `house_depth`, `road_width`, `bedroom_count`, `bathroom_count`) and the remaining eleven are text. Six categorical columns were converted to the categorical type, which reduced memory use from 6.533 GB to 5.334 GB. A conservative strategy was adopted: quality problems were flagged without removing rows from the full dataset, and a separate subset of priced listings was constructed afterwards (Section 4.2).

Missing values. Table 2 reports the missing values per variable. Five variables are complete, `area` is missing in 4 listings, and `price` is missing in 218,494 listings (6.24%). Several structural attributes have high missing rates, ranging from 68.96% for `house_direction` to 98.04% for `house_depth`. A missing `project_name` often indicates that the listing does not belong to a named development project.

Table 2: Missing values by variable in the raw data ($N = 3{,}500{,}744$).

Variable	Missing (n)	Missing (%)
`house_depth`	3,432,288	98.04
`balcony_direction`	2,933,342	83.79
`road_width`	2,830,873	80.86
`floor_count`	2,691,982	76.90
`project_name`	2,499,891	71.41
`house_direction`	2,414,214	68.96
`frontage_width`	1,726,424	49.32
`bathroom_count`	1,720,974	49.16
`bedroom_count`	1,600,361	45.71
`street_name`	1,376,860	39.33
`ward_name`	496,716	14.19
`price`	218,494	6.24
`district_name`	101,933	2.91
`area`	4	<0.01
`name`	0	0.00
`description`	0	0.00
`property_type_name`	0	0.00
`province_name`	0	0.00
`published_at`	0	0.00

The missing rate by property type (Figure 1) indicates that part of the missingness is related to the nature of the property. For apartments (*Căn hộ chung cư*), `floor_count`, `frontage_width`, `house_depth`, and `road_width` are missing in 100% of the listings; for land (*Đất*), `floor_count`, `bedroom_count`, `bathroom_count`, and `balcony_direction` are missing in 100%. The variable `project_name` is missing in 98% of houses (*Nhà*) and 93% of land listings, compared with 13% of apartments, whereas `price` is missing in 4% to 10% of the listings of each type. Rows with missing values were retained.

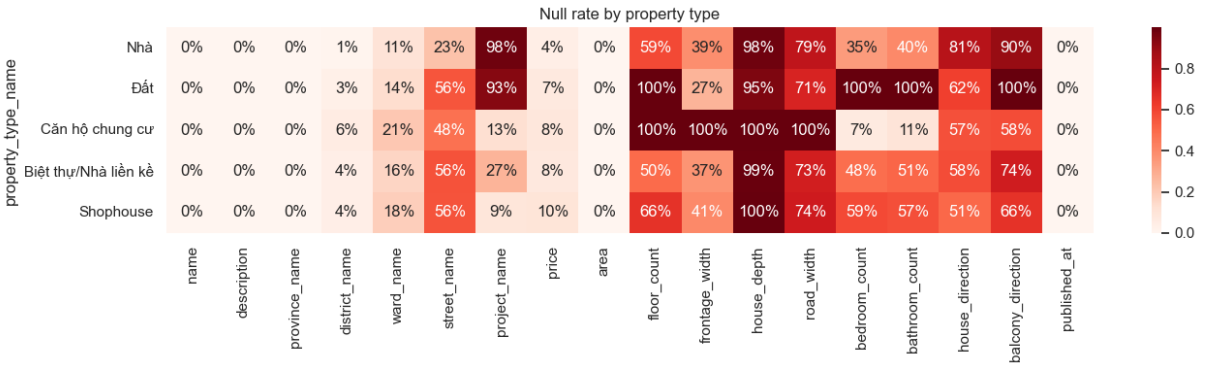


Figure 1: Null rate of each variable by property type.

Validity checks. There are 57,320 listings with `price` equal to zero and none with a negative price. No listing has a non-positive `area`, and `floor_count`, `bedroom_count`, and `bathroom_count` contain no negative values. Listings without a valid price or area represent 7.88% of the raw data. This share ranges from 5.2% (*Nhà*) to 13.6% (*Shophouse*) across the five property types, and from 4.1% (Khánh Hòa and Lâm Đồng) to 18.8% (Hòa Bình) among the 15 provinces with the most listings, so these listings are spread across groups and no single group accounts for most of them.

Temporal, geographic and direction variables. The variable `published_at` had no missing values, but 22,024 values could not be converted to a timestamp (invalid values were coerced to missing). The parsed values span 1 June 2025 to 30 March 2026. The data contain 63 provinces and 627 districts, and 15 district names appear in more than one province, so `district_name` should be used together with `province_name`. The direction variables are free text: besides single-direction values, the same direction is sometimes written in two notations (for example, *Đông Nam* and *Đông - Nam*), and a few entries contain several directions, each occurring at most five times. No standardisation was applied at this stage.

Duplicates. Listings were flagged as duplicates when they share the normalised `name`, `province_name`, `district_name`, `ward_name` (lower case, diacritics removed, *đ* mapped to *d*, whitespace collapsed), together with the same `area` and `price`, as an earlier listing; the first occurrence is kept. Under this definition, 865,362 listings (24.72%) are duplicates. The ten most repeated combinations occur between 60 and 121 times each, all in Hồ Chí Minh, and eight of the ten share the same title opening (*xuat canh ban gap*, normalised form).

Extreme prices. In the priced, de-duplicated data, the distribution of `price` on logarithmic axes (Figure 2) is concentrated between about 10^9 and 10^{10} VND, with isolated listings below 10^4 VND and above 10^{15} VND. The ten lowest prices are between 1,000 and 3,000 VND, and in several of them the title states a different price (for example, a listing recorded at 2,000 VND is titled as 1.6 billion VND). The ten highest range from about 4.49×10^{14} to 2.66×10^{16} VND, and some have very large areas (for example, 999,999 m^2). These observations suggest data entry errors, for example in the unit of the price. A range of $[10^8, 10^{12}]$ VND (100 million to 1,000 billion VND) was therefore examined: it would remove 2,574 listings (0.107%), and it was applied as a cleaning step (Section 4.2).

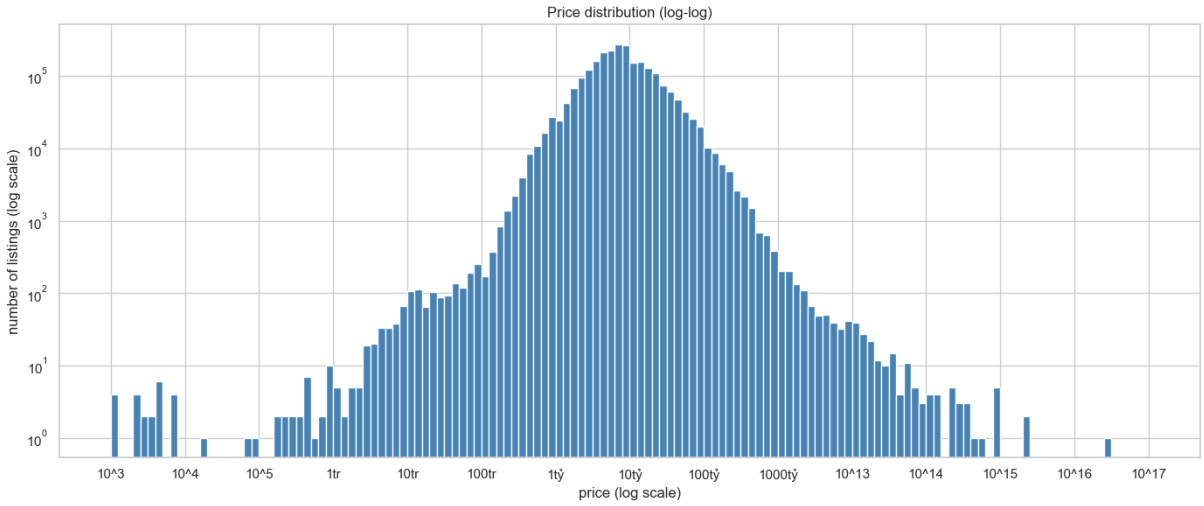


Figure 2: Distribution of asking price after removing invalid prices and duplicates (logarithmic scale on both axes).

Extreme values in other attributes. The numeric attributes and the price per square metre were inspected in the same data (Figure 3; $\log_{10}$ scale on the horizontal axis and logarithmic scale on the vertical axis). In each panel most listings lie in a narrow range, while sparse tails extend over several orders of magnitude. The maximum values are 82,000,000 m^2 for `area`, 255 for `floor_count`, 9,080,000 m for `frontage_width`, 15,380 m for `house_depth`, 18,000 m for `road_width`, 200 for `bedroom_count`, 168 for `bathroom_count`, and about 2.70×10^{13} VND per m^2 for `price_per_m2`. These values are not plausible for most properties. The separated bars for `floor_count`, `bedroom_count`, and `bathroom_count` reflect that these variables take integer values.

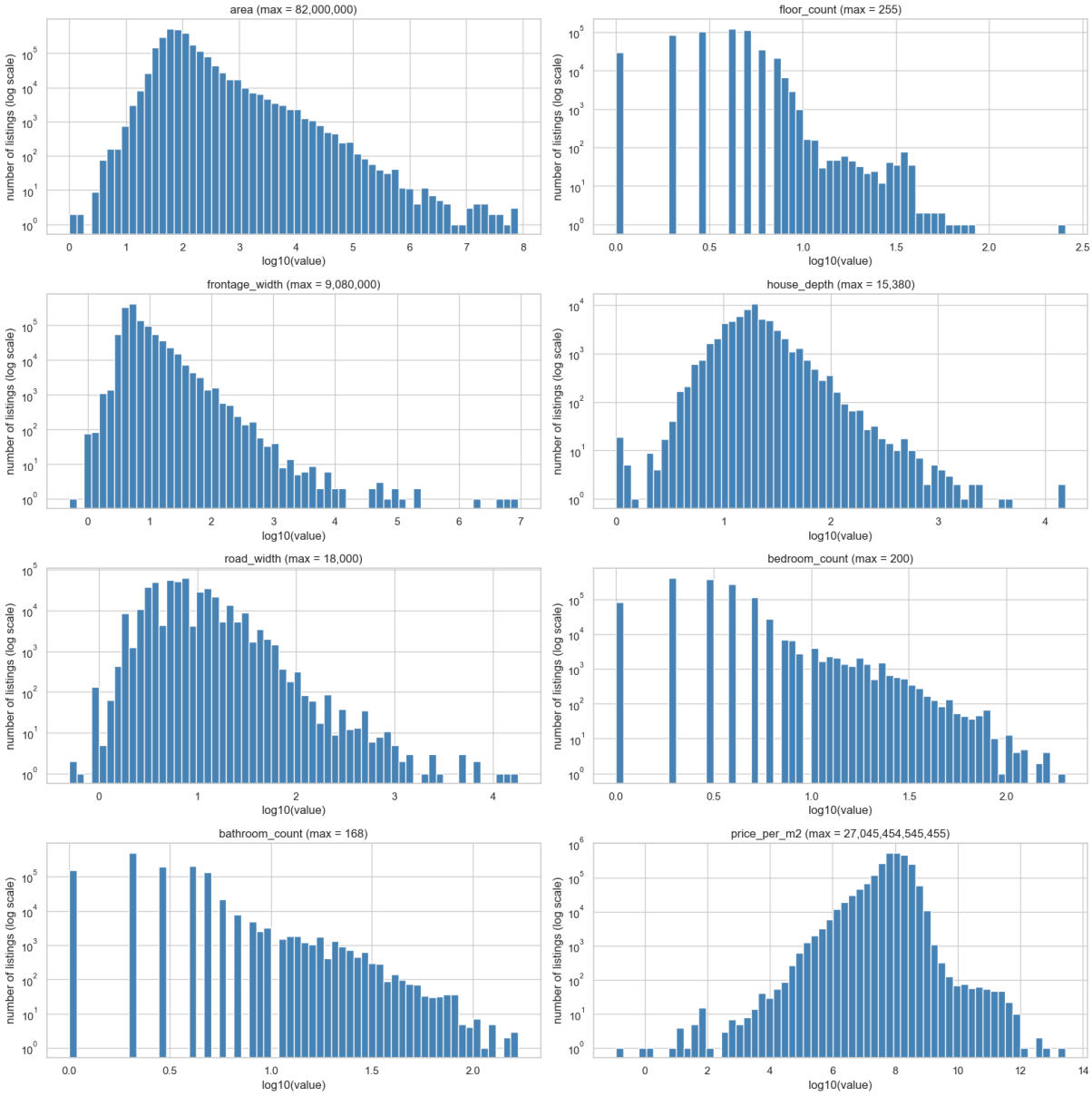


Figure 3: Distributions of the numeric attributes and of the price per square metre ($\log_{10}$ scale; vertical axis in logarithmic scale), before the plausibility limits are applied. The maximum of each variable is given in the panel title of the original figure.

4.2 Analytical Dataset

The analytical dataset (`df_priced`) was built in six steps, applied in the order of Table 4: (i) retaining listings with price > 0 and area > 0; (ii) removing the flagged duplicates; (iii) retaining prices within $[10^8, 10^{12}]$ VND; (iv) removing listings without a valid publication date; (v) applying plausibility limits to `area` and the other numeric attributes (Table 3); and (vi) removing listings with a price per square metre above 2 billion VND. In step (v), listings with an `area` outside its limits were removed, whereas for the other six attributes the out-of-range values were set to missing and the listings were retained. The number of values outside the limits was counted before step (v) was applied. The final dataset contains 2,397,001 listings.

Table 3: Plausibility limits applied to the numeric attributes.

Variable	Valid range	Values outside (n)	Treatment
`area` (m^2)	$[10, 100{,}000]$	876	Row removed
`floor_count`	$[1, 50]$	8	Set to missing
`frontage_width` (m)	$[1, 200]$	761	Set to missing
`house_depth` (m)	$[2, 500]$	70	Set to missing
`road_width` (m)	$[1, 100]$	424	Set to missing
`bedroom_count`	$[1, 50]$	340	Set to missing
`bathroom_count`	$[1, 50]$	267	Set to missing

Table 4: Number of listings after each filtering step.

Step	Rule	Listings (n)	Removed (n)
0	Raw data	3,500,744	—
1	Price > 0 and area > 0	3,224,930	275,814
2	Duplicates removed	2,415,372	809,558
3	Price within $[10^8, 10^{12}]$ VND	2,412,798	2,574
4	Valid publication date	2,398,343	14,455
5	Area within $[10, 100{,}000]$ m^2	2,397,467	876
6	Price per $m^2 \leq 2 \times 10^9$ VND (final)	2,397,001	466

The number removed in step 2 is the difference between the counts before and after the step.

4.3 Distributions and Relationships in the Analytical Dataset

Descriptive profile. Table 5 reports descriptive statistics of the numeric variables after the cleaning steps. `price` and `area` are available for all listings, whereas the other attributes are available for fewer (from 58,878 for `house_depth` to 1,304,397 for `bedroom_count`). The median price is 7.4 billion VND and the mean (1.49×10^{10} VND) is about twice as large, which indicates a right-skewed distribution; `area` behaves similarly (median 80 m^2, mean 217 m^2, standard deviation 1,619 m^2). The maximum of each attribute now equals the upper limit of Table 3 (for example, 100,000 m^2 for `area` and 200 m for `frontage_width`), so the tails are bounded.

Table 5: Descriptive statistics of numeric variables in the analytical dataset (N = 2,397,001).

Variable	Count	Mean	Std. dev.	Min	25%	50%	75%	Max
`price` (VND)	2,397,001	1.49×10^{10}	3.20×10^{10}	1.00×10^8	4.05×10^9	7.40×10^9	1.43×10^{10}	1.00×10^{12}
`area` (m^2)	2,397,001	217.38	1,619.32	10.00	56.00	80.00	120.00	100,000
`floor_count`	516,628	3.90	1.76	1	3	4	5	50
`frontage_width` (m)	1,208,300	7.54	8.97	1.00	4.20	5.00	7.50	200
`house_depth` (m)	58,878	22.57	20.84	2.00	13.41	18.75	24.64	500
`road_width` (m)	414,645	9.08	8.62	1.00	4.00	6.00	10.50	100
`bedroom_count`	1,304,397	3.30	2.34	1	2	3	4	50
`bathroom_count`	1,220,972	3.05	2.34	1	2	2	4	50

Distribution of price and area. On a base-10 logarithmic scale (Figure 4), the distribution of price is unimodal and approximately symmetric, with a mode at about 9.9 (roughly 8 billion VND) and values mainly between 8.5 and 11.5. The distribution of area is unimodal with a mode at about 1.9 to 2.0 (roughly 80 to 100 m^2) and a right tail that extends to about 5 (the upper limit of 100,000 m^2). Given the right skew shown in Table 5, this supports the use of a log transformation for price.

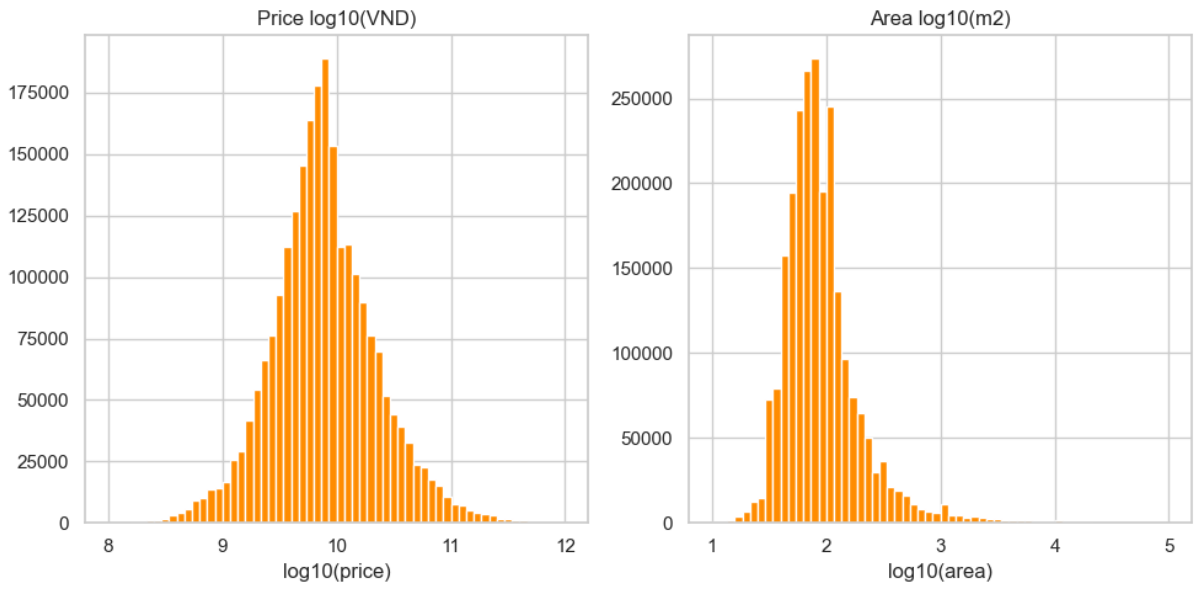


Figure 4: Histograms of $\log_{10}$(price) (left) and $\log_{10}$(area) (right).

Group coverage. Groups with fewer than 100 listings were considered difficult to use as features. All five property types exceed this threshold: houses (1,074,788), land (560,294), apartments (544,268), villas/townhouses (178,654), and shophouses (38,997). Of the 63 provinces, 60 exceed it. Listings are concentrated in Hồ Chí Minh (928,687) and Hà Nội (776,760), which together account for about 71% of the analytical dataset.

Area and price. A scatter plot on logarithmic axes, using a random sample of 20,000 listings (seed 42), is shown in Figure 5. The area axis spans 10 to 10^5 m^2, the limits applied in the cleaning. Most points lie at areas of roughly 30 to 300 m^2 and prices of roughly 10^9 to 10^{10} VND. The cloud shows an upward tendency, so larger areas are generally associated with higher prices, but the dispersion at a given area is wide, and points become sparse above 10^3 m^2. This plot is a visual indication only; the strength of the association was not quantified.

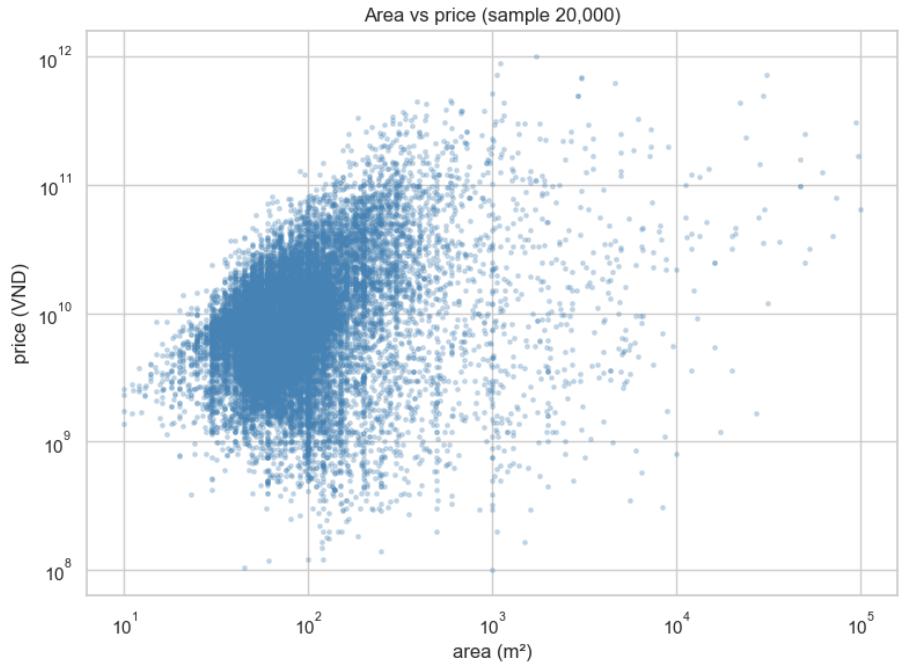


Figure 5: Area versus asking price on logarithmic axes (random sample of 20,000 listings).

Price per square metre by property type. The price per square metre (`price_per_m2`) is `price` divided by `area`. Its $\log_{10}$ distribution for each property type is shown in Figure 6, with the horizontal axis restricted to 6–10 (the data are capped at 2 billion VND per m^2, i.e. $\log_{10} \approx 9.3$). Houses and villas/townhouses are centred at higher values (modes at about 8.3 and 8.2, roughly 200 and 160 million VND per m^2) than apartments and land (modes at about 7.8). Land has the widest, left-skewed distribution, and apartments the narrowest. Shophouses show two local maxima, at about 7.8–7.9 and 8.2, which may indicate a mix of different kinds of listing (for example, sale and rental); this was not tested and remains a hypothesis.

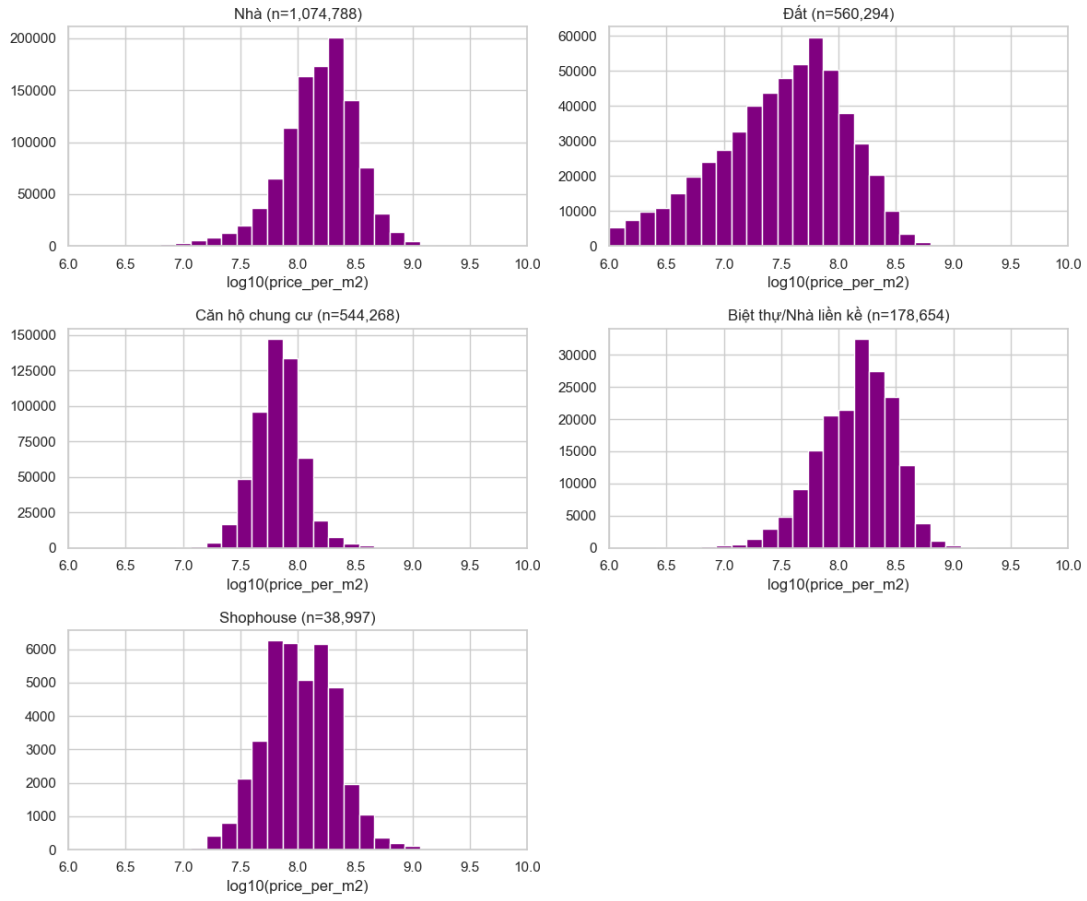


Figure 6: Distribution of $\log_{10}$(price per m^2) by property type (houses $n = 1{,}074{,}788$; land $n = 560{,}294$; apartments $n = 544{,}268$; villas/townhouses $n = 178{,}654$; shophouses $n = 38{,}997$).

4.4 Summary

Missing values are frequent for structural attributes and depend on the property type, 24.72% of the raw listings were flagged as duplicates, and 7.88% lack a valid price or area. Extreme values were found in price, area, the other physical attributes, and the price per square metre, and were handled with fixed limits; after all filtering steps, 2,397,001 listings remain. Price and area are right-skewed, and the log transformation of price gives an approximately symmetric distribution. All five property types and 60 of 63 provinces have sufficient listings, although listings are concentrated in two provinces. Area and price show a positive but widely dispersed relationship, and the price per square metre differs by property type.

References

[Tin26] TiniX AI. *Tinix Vietnam Real Estate Listings 2025–2026*. Hugging Face. 2026. URL: https://huggingface.co/datasets/tinixai/vietnam-real-estates (visited on 04/10/2026).